Unit 6
Digital Literacy

Find, collect, judge, and present information without fooling yourself

QUALITATIVE VS QUANTITATIVE | PRIMARY VS SECONDARY | SEARCH OPERATORS | RESEARCH ETHICS

Digital literacy is not “I can use a phone”. It is using a phone without being used by it.

0. Literacy for the information age

Digital literacy is the mix of technical, thinking, and social skills that lets a person find information, organise it, turn it into something useful, and share it responsibly. Access to a device is not enough. The Information Age dumps more pages on you in a day than a 20th-century library did in a month; the scarce skill is judgement.



Figure 1. Five verbs of digital literacy. Skipping “judge” is how rumours become assignments.

1. Two kinds of data

QUALITATIVE

Descriptive. Opinions, reasons, observed behaviour. “Students say the LMS login fails during exams.” Helps you understand *why*.

QUANTITATIVE

Numerical. Counts, marks, hours online. “62% of the class uses the internet more than 4 hours a day.” Helps you compare and chart.

A decent school project almost always needs both: hours on the phone (quantitative) and the sentence “I check notifications between every paragraph” (qualitative).

2. How you collect it

Table 1. Data-collection strategies

Strategy	You do	Best for
Interview	Ask a person, follow interesting answers	Depth, reasons
Survey	Same questions to many people (Forms)	Breadth, percentages
Observation	Watch a lab or classroom without interrupting	What people actually do
Prototype	Show a rough design, watch the reaction	Product / HCI ideas (see XII Unit 1)
Simulation	Model a system and run it	Dangerous or expensive real trials

3. Primary versus secondary

Table 2. Who collected it, and for whom?

	Primary	Secondary
Origin	You collected it for *this* question	Someone else already collected it
Control	High — you wrote the questions	Low — you inherit their choices
Example	Survey of your own Class XI	PBS report, a journal paper, a government PDF

4. Designing a data-collection approach

1. Write one clear **research question** (not a topic). Example: “How does daily mobile-phone use affect study habits of Class XI students in this college?”
2. Define the **population** and a realistic **sample**.
3. Choose methods (survey + two interviews is a strong school mix).
4. Draft fair, non-leading questions. Mix closed (numbers) and open (reasons).
5. Pilot on three classmates; fix confusing wording.
6. Ask **permission**. Do not collect names you do not need. That is research ethics, not decoration.
7. Collect, back up, clean.
8. Analyse and present. Then say what you still do *not* know.

5. Search like a researcher, not like a tourist

A lazy query dumps blogs. Operators carve the pile.

```
"internet usage in Pakistan" filetype:pdf
site:.edu  "study habits"  smartphone
"mobile phone use" "study habits"
```

Table 3. Operators worth memorising

Operator	Job
`"exact phrase"`	Keep those words together
`filetype:pdf`	Prefer reports over slideshows
`site:edu` or `site:gov.pk`	Stay inside a domain family
`OR` / `-`	Widen / exclude a word

Source checklist: named author or institution? date? citations? evidence or only adjectives? other side mentioned?

6. Presenting without drowning the reader

SPREADSHEET
Excel / Google Sheets: tidy columns, formulas, charts. They do not “make the research correct”; they organise what you already collected.

PRESENTATION
One claim per slide, a chart, a short sentence. Not a pasted essay.

INFOGRAPHIC
A single visual argument for a corridor wall or a report cover.

REPORT / ARTEFACT
Question, method, findings, limits, references. The artefact’s job is to communicate results, not to decorate.

7. Integrated case — phone use and study habits

The textbook walks a full mini-project: advanced search, source checklist, a Google/Microsoft Form, Sheets charts, and a digital artefact. Quantitative layer: hours per day, percentage who report distraction. Qualitative layer: why they pick up the phone. Ethics: permission, no unnecessary names, backups.

ETHICS & HONESTY

Copying words, images, or ideas without credit is **plagiarism**. Copyright is the legal shield; a citation is how you stay on the right side of it. Asking permission before a survey is part of the method, not an afterthought.

Self-check (exam style)

1. Which query is best for a reliable PDF on internet use in Pakistan? (a) `internet usage Pakistan` (b) `best internet social media` (c) `"internet usage in Pakistan" filetype:pdf` (d) `Pakistan internet all websites latest`
2. Survey responses from your own class are _____ data.
3. Name the error: pasting a blog paragraph into your report with no quote marks and no reference.

Answers. (1) (c). (2) Primary. (3) Plagiarism.

Recap

1. Digital literacy = find + collect + judge + present + cite.
2. Qualitative explains; quantitative measures. Use both.
3. Interviews, surveys, observation, prototypes, simulations.
4. Primary = you collected it; secondary = you reused it.
5. A research question focuses the whole design. Permission is ethics.
6. Search operators: quotes, `filetype:`, `site:`.
7. Spreadsheets, slides, infographics, reports — clarity over decoration. Never plagiarise.